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Systems, devices, and methods for marking and/or reinforcing fenestrations in prosthetic implants

Abstract

Devices, systems, and methods for marking and/or reinforcing fenestrations in grafts are disclosed herein. In some embodiments, a radiopaque marker for a graft is provided. The marker can be secured to the graft in the area near a fenestration such that the marker is visible via radiographic imaging. The radiopaque marker can be in the form of a radiopaque thread, a radiopaque bead, a radiopaque additive, or a radiopaque adhesive. In some embodiments, the radiopaque marker is in the form of a circular disc shaped and sized to surround a fenestration, the circular disc being formed of a radiopaque material. The radiopaque marker can be configured to or can be attached to a reinforcement member configured to reinforce a fenestration such that one or more edges of the fenestration are protected and/or to aid in engagement with and sealing to a mating stent.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/203,873, filed Nov. 29, 2018, which is a continuation of International Patent Application No. PCT/US2017/037157, filed Jun. 13, 2017, which claims priority to and the benefit of U.S. Patent Application No. 62/349,287, filed Jun. 13, 2016, entitled “Systems, Devices, and Methods for Marking and/or Reinforcing Fenestrations in Prosthetic Implants,” the disclosures of each of which are incorporated herein by reference in their entirety.

BACKGROUND

(1) Embodiments described herein relate generally to devices, systems and methods for marking and/or reinforcing fenestrations in grafts, such as, for example, aortic stent grafts.

(2) Aneurysms generally involve the abnormal swelling or dilation of a blood vessel such as an artery. The wall of the abnormally dilated blood vessel is typically weakened and susceptible to rupture. For example, an abdominal aortic aneurysm (AAA) is a common type of aneurysm that poses a serious health threat. A common way to treat AAA and other types of aneurysm is to place an endovascular stent graft such that the stent graft spans across and extends beyond the proximal and distal ends of the diseased portion of the vasculature. The stent graft is designed to reline the diseased vasculature, providing an alternate blood conduit that isolates the aneurysm from the high pressure flow of blood, thereby reducing or eliminating the risk of rupture.

(3) Minimally invasive endovascular repair using stent grafts is often preferred to avoid the risks associated with traditional open surgical repair. However, these stent grafts can only be used when the graft can be placed in a stable position without covering major branch vessels. In the cases of juxtarenal aneurysm where the dilation extends up to but does not involve the renal arteries, the proximal portion of the stent graft needs to be secured to the aortic wall above the renal arteries, thereby blocking the openings to the renal arteries. Thus, patients with juxtarenal aneurysms, which represent a significant proportion of abdominal aortic aneurysm cases, are typically excluded from endovascular treatment.

(4) To allow for endovascular repair of a wider range of cases, openings are sometimes created during manufacturing or cut by surgeons in the stent graft body to accommodate specific branch vessel origins, a process known as “fenestration.” Thus, for example, in treating juxtarenal aneurysms, the fenestrations or openings of the stent grafts are to be aligned with the renal arteries. Traditionally, the fenestration process involves measurements based on medical images (such as CT scans) of the vessel origins. Longitudinal distances may be measured, and relative angular locations may be estimated from a reference point.

(5) However, these manual measurements may take a substantial amount of time and effort, particularly when multiple branch vessels must be accommodated. For example, in abdominal

aortic aneurysms, fenestrations may be required for both left and right renal arteries, the superior mesenteric artery (SMA), and the celiac artery. In addition, approximations of the placement of the branch openings could lead to errors in the placement of the openings compared to the true branch vessel origins. In some cases, openings may be erroneously placed over stent struts. In operating room conditions, surgeons often need to cut fenestrations in the stent body quickly. Additionally, there are challenges associated with cutting graft material both when cut by surgeons in operating room conditions and when fenestrations are created during manufacturing of a graft. Typical graft material is flexible and shifts in response to being pressed on with a cutting tool. Therefore, there is a need for a simple yet accurate and cost-effective way to create fenestrations in stent grafts. Moreover, there is a need for reinforcement of portions of grafts surrounding the fenestrations and for marking the fenestrations such that the fenestrations can be easily located during delivery and/or while the graft is in use.

SUMMARY

(6) Devices, systems, and methods for marking and/or reinforcing fenestrations in grafts are disclosed herein. In some embodiments, a radiopaque marker for a graft is provided. The marker can be secured to the graft in the area near a fenestration such that the marker is visible via radiographic imaging. In some embodiments, the radiopaque marker is in the form of a radiopaque thread, a radiopaque bead, a radiopaque additive, or a radiopaque adhesive. In some embodiments, the radiopaque marker is in the form of a circular disc shaped and sized to surround a fenestration, the circular disc being formed of a radiopaque material. In some embodiments, the radiopaque marker can be configured as or can be attached to a reinforcement member configured to reinforce a fenestration such that one or more edges of the fenestration are protected and/or aid in engagement with and sealing to a mating stent.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic illustration of a template, according to an embodiment.
- (2) FIG. 2 is a schematic illustration of a patch, according to an embodiment.
- (3) FIG. 3 is a perspective view of a graft in a first configuration, according to an embodiment.
- (4) FIG. 4 is an enlarged perspective view of the graft of FIG. 3 in a second configuration, according to an embodiment.
- (5) FIG. 5 is an enlarged perspective view of the graft of FIG. 3 in a third configuration, according to an embodiment.
- (6) FIG. 6 is an enlarged perspective view of the graft of FIG. 3 in a fourth configuration, according to an embodiment.
- (7) FIG. 7 is a front view of a reinforcing member attached to a graft, according to an embodiment.
- (8) FIG. 8 is a schematic illustration of a perspective view of a reinforcing member, according to an embodiment.
- (9) FIG. 9A is a schematic cross-sectional illustration of the reinforcing member of FIG. 8 in a first configuration.
- (10) FIG. 9B is a schematic cross-sectional illustration of the reinforcing member of FIG. 8 in a second configuration.
- (11) FIG. 9C is a schematic cross-sectional illustration of the reinforcing member of FIG. 8 in a third configuration.
- (12) FIG. 10 is a side view of a reinforcing member attached to a graft, according to an embodiment.
- (13) FIG. 11 is a side view of a reinforcing member attached to a graft, according to an embodiment.

- (14) FIG. 12 is a perspective view of a reinforcing member, according to an embodiment.
- (15) FIG. 13 is a front view of a reinforcing member attached to a graft, according to an embodiment.
- (16) FIG. 14 is a front view of a reinforcing member attached to a graft, according to an embodiment.
- (17) FIG. 15 is a side view of a reinforcing member attached to a graft, according to an embodiment.
- (18) FIG. 16 is a perspective view of a reinforcing member, according to an embodiment.
- (19) FIG. 17 is a perspective view of a reinforcing member, according to an embodiment.
- (20) FIG. 18 is a schematic illustration of a reinforcing member, according to an embodiment.
- (21) FIG. 19A is a perspective view of a reinforcing member, according to an embodiment.
- (22) FIG. 19B is a perspective view of an outer element of the reinforcing member of FIG. 19A.
- (23) FIG. 19C is a perspective view of an inner element of the reinforcing member of FIG. 19A.
- (24) FIG. 20 is a flow chart of a method, according to an embodiment.

DETAILED DESCRIPTION

- (25) Devices, systems, and methods for marking and/or reinforcing fenestrations in grafts are disclosed herein. In some embodiments, an apparatus includes a member and at least one radiopaque element. The member can be configured to be secured to a patient-specific prosthetic such that the member surrounds a fenestration defined by the prosthetic. The fenestration can correspond to a location of a branch blood vessel in a portion of a patient's blood vessel. The at least one radiopaque element can be configured to indicate the location of the fenestration via radiographic imaging.
- (26) In some embodiments, a radiopaque marker for a graft is provided. The marker can be secured to the graft in the area near a fenestration such that the marker is visible via radiographic imaging. In some embodiments, the radiopaque marker is in the form of and/or includes a radiopaque thread, a radiopaque bead, a radiopaque additive, a radiopaque wire or coil, a radiopaque powder embedded in another substrate, and/or a radiopaque adhesive. In some embodiments, the radiopaque marker is in the form of and/or includes a circular disc shaped and sized to surround a fenestration, the circular disc being formed of a radiopaque material.
- (27) In some embodiments, a marker template for a graft is provided. The fenestration template can include one or more openings corresponding to one or more desired marker locations on the graft. The fenestration template can be coupled to the graft such that marker elements can be applied to the graft via the openings. In some embodiments, marker elements can be attached to the fenestration template and transferred to the graft when the fenestration template is coupled to the graft.
- (28) In some embodiments, a reinforcing member (also referred to herein as a patch or a grommet) for a graft is provided. The reinforcing member can include radiopaque markers, be formed of a radiopaque material, and/or be embedded with a radiopaque material. The reinforcing member can be configured and applied to the graft such that the fenestration is reinforced and/or protected. For example, the reinforcing member can prevent fraying of the edge of a fenestration of the graft. In some embodiments, the reinforcing member can aid in engagement and sealing between the graft and another mating stent. In some embodiments, the reinforcement member can be formed as a patch configured to be delivered to a graft to reinforce the area surrounding the fenestration and mark the location of the fenestration. In some embodiments, the reinforcement member can be formed as a grommet configured to be delivered to a graft such that the grommet is secured within a fenestration of the graft. The grommet can reinforce the area of the graft surrounding the fenestration and mark the location of the fenestration.
- (29) As used in this specification, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, the term “a member” is intended to mean a single member or a combination of members, “a material” is intended to mean one or more

materials, or a combination thereof.

(30) As used herein, the words “proximal” and “distal” refer to a direction closer to and away from, respectively, an operator of, for example, a medical device. Thus, for example, the end of the medical device contacting the patient's body would be the distal end of the medical device, while the end opposite the distal end would be the proximal end of the medical device. Similarly, when a device such as an endovascular stent graft is disposed within a portion of the patient, the end of the device closer to the patient's heart would be the proximal end, while the end opposite the proximal end would be the distal end. In other words, the proximal end of such a device can be upstream of the distal end of the device.

(31) As used herein, “reinforced” and variations of “reinforced” (e.g. reinforcing, reinforcement, reinforce) means strengthened or supported such that an edge of a material is prevented from fraying, such that the shape of a portion of a material is maintained, and/or such that engagement and sealing with another material is improved.

(32) The embodiments described herein can be formed or constructed of one or more biocompatible materials. Examples of suitable biocompatible materials include metals, ceramics, or polymers. Examples of suitable metals include pharmaceutical grade stainless steel, gold, titanium, nickel, iron, platinum, tin, chromium, copper, tantalum, and/or alloys thereof. Examples of polymers include nylons, polyesters, polycarbonates, polyacrylates, polymers of ethylene-vinyl acetates and other acyl substituted cellulose acetates, non-degradable polyurethanes, polystyrenes, polyvinyl chloride, polyvinyl fluoride, poly(vinyl imidazole), chlorosulphonate polyolefins, polyethylene oxide, polyethylene terephthalate (PET), polytetrafluoroethylene (PTFE), and/or blends and copolymers thereof.

(33) The embodiments and methods described herein can be used to form a patient-specific prosthetic device and/or to facilitate the function and/or the integration of the prosthetic device within a portion of a patient. For example, in some embodiments, the devices and/or methods described herein can be used in conjunction with and/or can otherwise include endovascular repair using stent grafts. Although the embodiments are shown and described herein as being used, for example, to facilitate endovascular repair, in other embodiments, any of the devices and/or methods described herein can be used to facilitate treatment of any portion of a patient. For example, the devices and methods described herein can form and/or can facilitate the integration of any suitable implant, prosthesis, device, mechanism, machine, and/or the like within a portion of the body of a patient such as the patient's vascular system, nervous system, muscular-skeletal system, etc. Therefore, while the embodiments are shown and described herein as being used in the endovascular repair of an abdominal aortic aneurysm, they are presented by way of example and are not limited thereto.

(34) Some of the devices and/or methods described herein can be used in minimally invasive treatment techniques such as endovascular repair using stent grafts. Such repair techniques are generally preferred over traditional open surgical repair and often result in reduced morbidity or mortality rates. In some instances, however, the arrangement of the diseased vasculature can result in a need to alter a portion of the stent graft prior to insertion into the body. For example, in an endovascular repair of an abdominal aortic aneurysm, the aneurysm can be situated adjacent to and/or directly distal to normally functioning vessels branching from a portion of the aorta. In order to reline the aneurysm with the stent graft, surgeons often cut openings in the stent graft fabric to accommodate specific branch vessel origins, a process known as “fenestration.” Specifically, in treating juxtarenal aneurysms, for instance, the fenestrations or openings of the stent grafts can correspond to a size, shape, and/or relative position of, inter alia, the left and right renal arteries, the superior mesenteric artery (SMA), and/or the celiac artery.

(35) Traditionally, the fenestration process involves measurements based on medical images (such as CT scans) of the vessel origins. For example, in some instances, longitudinal distances of branch vessels can be measured and relative angular locations of the branch vessels can be estimated

and/or calculated from a reference point. Based on these measurements and/or calculations, a surgeon or manufacturer can mark and cut the stent fabric of a stent graft to define one or more fenestrations. The fenestrated stent graft can then be positioned within the diseased vasculature (e.g., via an endovascular procedure) and oriented to substantially align the fenestrations with openings of the corresponding branch vessels.

(36) In various embodiments, a fenestration can be created in a prosthetic implant, such as a stent graft, using any suitable method. For example, fenestrations can be created using a fenestration template manufactured using any suitable technologies such as 3-D printing or additive prototyping/manufacturing technologies, subtractive manufacturing techniques, 2-D printing, and the like or a combination thereof. In some embodiments, the fenestration templates are generated for patient-specific anatomy, for example, based on patient specific image data. Examples of such fenestration templates and the generation of such fenestration templates are described in U.S. Patent Publication No. 2013/0296998, filed May 1, 2013, entitled “Fenestration Template For Endovascular Repair of Aortic Aneurysms”; U.S. Pat. No. 9,629,686, issued Apr. 25, 2017 and titled “Devices and Methods for Anatomic Mapping for Prosthetic Implants”; and U.S. Pat. No. 9,629,705, issued Apr. 25, 2017 and titled “Devices and Methods for Anatomic Mapping for Prosthetic Implants”; the entire disclosures of which are incorporated herein by reference.

(37) In some embodiments, a fenestration can be created in a stent graft using any suitable method. For example, similarly as described with reference to FIGS. 3-6 below, a number of cuts can be created to segment the graft into a number of triangle or pie slice-shaped flap portions. The flap portions can be pulled or folded into a folded configuration such that the flap portions can be attached to the outer surface of the graft, resulting in the creation of a fenestration. One or more suture threads can be used to attach or reinforce the attachment of the flap portions to the graft in the folded configuration. The suture thread can include a material with radiopaque properties, such as, for example, gold, for visualization of the suture thread using radiographic imaging.

Visualization of the suture thread can help a user to identify the location and/or orientation of the graft. Alternatively or in addition to the suture thread, fasteners such as, for example, staples, rivets, micro-rivets, adhesives, and/or welding can be used to secure the flap portions to the graft in the folded configuration. The fasteners can include or be formed of material with radiopaque properties such that the fasteners are visible using radiographic imaging.

(38) In some embodiments, one or more radiopaque beads can be attached to the graft. The beads can be perforated such that each bead can receive a thread for attachment (e.g., sewing) of the beads to the graft. In some embodiments, the beads can be solid. Alternatively or in addition to being sewn to the graft, the beads can be coated with an adhesive material to bond the beads to the graft. For example, the adhesive can be a heat-activated or low melting temperature adhesive or a pressure sensitive adhesive. In some embodiments, the beads themselves can be made from a low melting temperature material that can bond to the graft directly. Any suitable number of beads can be attached to the graft.

(39) In some embodiments, a non-discrete marker can be used to identify the location of and/or reinforce fenestrations in the graft. For example, radiopaque glue can be applied to the graft. In some embodiments, prior to creating a fenestration, the glue can be applied to an area of the graft near a portion intended to be fenestrated. For example, the glue can be applied as a circular band surrounding the area intended for fenestration. After application of the glue, the portion inside of the circular band can be cut to create flap portions as described above. To create the fenestration, the flap portions can be folded into contact with the glue.

(40) In some embodiments, a circular marker, such as a ring or washer-shaped marker, can include a radiopaque material, such as a radiopaque fiber or a radiopaque powder, and be attached to the graft. In some embodiments, the radiopaque fiber or the radiopaque powder can be uniformly distributed throughout the circular marker. Adhesives, such as pressure sensitive adhesives and/or silicone adhesives, can be used to secure the circular marker to the graft. In some embodiments, the

circular marker can be secured to the graft via a thermal process. For example, the circular marker can be secured to the graft via an adhesive that can be a heat-activated or a low melting temperature adhesive. Alternatively or additionally, fasteners, such as, for example, staples, rivets, and micro-rivets, can be used to secure the marker to the graft. In addition, as described above, the fasteners can include a material with radiopaque properties such that both the marker and the fasteners are visible using radiographic imaging. Although described as a circular marker, the marker can be any suitable shape, such as ovular, flower-shaped, star-shaped, or rectangular.

(41) In some embodiments, a marker template can be used to aid in positioning radiopaque elements. Similar to the fenestration templates described above, the marker template can be 3-D printed. In some embodiments, the features of a fenestration template and a marker template can be combined to form a combined fenestration and marker template configured to aid in positioning one or more fenestrations and in applying one or more markers.

(42) In some embodiments, a marker template or a combined fenestration and marker template can be formed (e.g., printed) such that markers are incorporated into the template. Additionally or alternatively, the marker template or the combined fenestration and marker template can define apertures configured to receive the markers. For example, FIG. 1 is a schematic illustration of a combined fenestration and marker template **100**. As shown in FIG. 1, the template **100** includes a fenestration aperture **110** and a first marking aperture **120A**, a second marking aperture **120B**, a third marking aperture **120C**, and a fourth marking aperture **120D** (collectively referred to herein as “marking apertures **120**”). The template **100** can be coupled to a graft such that the fenestration aperture **110** is aligned with a desired fenestration location and each of the marking apertures **120** are aligned with a desired graft marker location. A cutting tool (such as any of the cutting tools described herein) can be used to cut out the portions of the graft aligned with the fenestration aperture **110**. In some embodiments where a contact cutting tool is used (e.g., a cutting tool with a sharp blade or a cautery device), the cutting tool can be inserted through the fenestration aperture **110** into cutting contact with the graft to cut out the portion of the graft aligned with the fenestration aperture **110**. In some embodiments where a non-contact cutting tool is used (e.g., an air knife, a waterjet, or a plasma torch), the cutting tool can be aligned with the fenestration aperture **110** such that the cutting mechanism uses the fenestration aperture **110** as a guide or an outline for cutting the graft. Additionally, a marking tool can be used to apply a radiopaque element (i.e., a marker), such as any of the radiopaque elements described herein, to the portions of the graft aligned with the marking apertures **120**. For example, the marking apertures **120** are shaped and sized to allow markers, such as the radiopaque beads described above, to be sewn into the graft through the marking apertures **120** or pop riveted to the graft through the marking apertures **120**. The template **100** can then be removed from the graft.

(43) In some embodiments, a fenestration in a graft can be created, before or after the application of any of the reinforcing members described herein, using, for example, a mechanical cutting means (e.g., a sharp blade) or heat application. Additionally, in some embodiments, the cutting tool or another tool can be used to apply heat to seal the edges of the graft. In some embodiments, the cutting tool used to create any of the fenestrations described herein can be harpoon-shaped or U-shaped (i.e., hook-shaped), allowing for the material of the graft to be supported and pulled toward the user during a pull stroke of the user. The cutting tools described herein can be used to create any suitable number, shape, and size of cuts and/or fenestrations.

(44) In some embodiments, one or more markers, such as the radiopaque beads or circular marker described above, can be disposed on an inner surface of the template. A graft can be positioned within the template, and a balloon can be disposed within the graft. The balloon can be inflated such that the balloon presses the graft against the inner surface of the template. In some embodiments, the markers can be automatically transferred from the template to the graft via an adhesive coating on the markers. In some embodiments, the markers can be secured to the graft via a fastener. In some embodiments, one or more markers can be disposed on the outer surface of a

template and the template can be positioned within a graft. External pressure can be applied to the graft such that the one or more markers on the outer surface of the template can transfer to the graft via adhesive or the application of fasteners.

(45) In some embodiments, markers (e.g., radiopaque beads) can be secured to a graft, such as an endograft, prior to the graft being cut or fenestrated, to aid in a cutting operation. For example, one or more markers can be sewn into a graft with individual threads. Tension can be maintained on the threads such that the material of the graft can be held taut to aid in cutting the graft. The graft can be cut such that flap portions are created, similarly as described above. The flap portions can be folded and secured to the outer surface of the graft such that a fenestration is defined and reinforced.

(46) In some embodiments, a graft, such as an endograft, can be cut such that flap portions are created, similarly as described above. One or more markers can be placed on or near the flap portions. Each flap portion can be folded such that each flap portion sandwiches at least one of the markers between the flap portion and an outer surface of the graft. The flap portions can then be secured in position using, for example, one or more sutures, adhesive, and/or heat-bonding, thereby securing the radiopaque markers in position.

(47) In some embodiments, a cutting and marking tool can be used to create the fenestration and apply the markers. The cutting and marking tool can be used to create a fenestration similarly to any of the cutting tools described herein. For example, the cutting and marking tool can include a cutting portion and a piercing portion. In some embodiments, the piercing portion can be disposed on an end of the cutting portion. In some embodiments, the piercing portion can be a separate component of the cutting tool than the cutting portion. The piercing portion can be used to create a pilot hole in a graft. For example, the piercing portion can be pushed distally into piercing contact with the graft in an area where a fenestration is desired until the piercing portion has created a pilot hole in the graft. The cutting portion or remainder of the cutting portion can be moved through the pilot hole and into the interior of the graft. Once the cutting portion is on the interior of the graft, the cutting portion can be pulled proximally such that it creates a cut in the graft as it is being pulled proximally and away from the interior of the graft. The cutting and marking tool can also include a marking portion. In some embodiments, the marking portion can deliver a marker to the graft and/or secure a marker to the graft using, for example, suture thread, fasteners, or adhesive.

(48) In some embodiments, a reinforcement member is in the form of a flexible or compliant patch. The flexible or compliant patch can be applied to a graft, such as an endograft, to reinforce and/or mark a fenestration defined by the graft. For example, the patch can be coupled to the area of the intended fenestration to reinforce the fenestration (e.g., prevent fraying) and/or to aid in stiffening the graft for cutting. In some embodiments, the patch can include radiopaque materials. For example, the patch can include radiopaque materials distributed substantially uniformly throughout the patch material. The patch can be formed as a preassembled membrane. In some embodiments, the patch can include one or more radiopaque markers. The radiopaque markers can be embedded in the patch material or secured to the surface of the patch. The patch can be applied to the graft either before or after the fenestration is created. In some embodiments, the patch can be formed as a radiopaque donut-shaped marker. The donut-shaped marker can be flexible and can be formed of a radiopaque membrane material.

(49) The patch can be attached to the graft using any suitable means. In some embodiments, the patch can be sewn to the graft. For example, the patch can be attached via sewing at a number of locations (e.g., four or six). In some implementations, the patch can be attached to the graft via a method in which needles are pre-loaded with sutures and attached to the patch such that all the needles can be activated simultaneously to deliver the sutures through the patch and graft material. In some embodiments, the patch can be secured to the graft with an adhesive, such as a pressure-sensitive adhesive, cyanoacrylate, or a silicone adhesive. The patch can also be secured to the adhesive via heat bonding. In some embodiments, the patch can be heat bonded to the graft. For

example, the patch and the graft can both be formed of DACRON® (i.e., polyethylene terephthalate) such that the application of thermal energy creates a DACRON® to DACRON® bond. Alternatively, the patch can be formed of a material with a lower melting point than DACRON® (i.e., such that a temperature differential exists) such that the application of thermal energy allows the patch material to flow within the fibers of the DACRON® material for securement. In some embodiments, the patch can be formed of polyurethane and the graft can be formed of polyethylene terephthalate such that the application of thermal energy can bond the patch to the graft. Additionally, a fenestration in the graft can be created via the application of thermal energy simultaneously, before, or after the bonding of the patch to the graft. In some embodiments, the patch can be secured to the graft via fasteners such as, for example, staples or rivets. In such embodiments, the fasteners can include radiopaque materials.

(50) The patch can be formed of any suitable material and in any suitable shape or configuration. For example, the patch can be formed of a radiopaque fabric or of any flexible material with a radiopaque material embedded in the flexible material. Additionally or alternatively, in some embodiments, radiopaque markers, such as the radiopaque beads described above, can be embedded in the patch. FIG. 2 is a schematic illustration of a patch **200**. As shown in FIG. 2, the patch **200** can be circular. Markers, such as first marker **230A**, second marker **230B**, and third marker **230C** (collectively referred to herein as “markers **230**”), can be arranged on the patch **200** in a predetermined pattern, such as a bullseye or a star pattern. In some embodiments, the markers **230** can be used as a cutting template for creating a fenestration in a graft, such as an endograft. In such embodiments, some of the markers **230** may be discarded after cutting the graft during a chard removal step.

(51) In some embodiments, the patch can be formed as a ring. Radiopaque markers can be embedded within the ring. For example, the ring-shaped patch can be formed of silicone or thermoplastic elastomer and molded into a ring shape. Radiopaque markers, such as, for example, tungsten, can be embedded within the outer surface of the ring. In some embodiments, radiopaque markers can be attached to the outer surface of the ring. In some embodiments, the patch can be formed as a circular donut. The circular donut-shaped patch can include a radiopaque material. For example, the circular donut-shaped patch can be formed of foil, radiopaque fiber, or metalized film.

(52) FIGS. 3-6 are various views of an example of a system including a graft **361** and a reinforcing member **340** (also referred to herein as “a reinforcement and marking patch”) that can be coupled to the area of the intended fenestration of the graft **361** to aid in stiffening the graft **361** for cutting, to prevent fraying, and/or to mark the location of the fenestrations. The reinforcement and marking patch **340** can be coupled to the graft **361** before or after a cutting operation. For example, FIG. 3 is a perspective view of a graft **361**. The graft **361** can be, for example, an endograft. As shown, the graft **361** includes a pilot hole **365**, cuts **367**, and flap portions **369**. Although the graft **361** is shown as having been cut such that it includes eight cuts **367** and eight flap portions **369**, the graft **361** can include any suitable number of cuts or flap portions. Additionally, the pilot hole **365** and the cuts **367** can be created through any suitable method described herein.

(53) After the flap portions **369** have been created in the side of the graft **361**, the reinforcement and marking patch **340** can be coupled to the area surrounding the flap portions **369**. For example, FIG. 4 is a close-up view of a portion of the graft **361** with the patch **340** coupled to the graft **361**. The patch **340** can be ring or donut-shaped and can be coupled to the graft **361** via any suitable attachment means, such as, for example, adhesive (e.g., pressure sensitive adhesive or silicone adhesive), sutures, or welding (i.e. melting) of the patch **340** to the graft **361**. The patch **340** can be arranged relative to the flap portions **369** such that the edges of the cuts **367** are aligned with an internal edge of the patch **340**. Although the patch **340** is described as being coupled to the graft **361** after the cuts **367** are created, the patch **340** can be coupled before the creation of the cuts **367** to strengthen the material of the graft **361** in the area of the intended cuts **367**. Pre-cut application of the patch **340** can increase the rigidity and/or tautness of the graft **361** in the area of the graft **361**

encircled by the reinforcement patch **340** such that the graft **361** is easier to cut. Furthermore, the patch **340** can include radiopaque elements or be formed of a radiopaque material, similar to any of the patches described herein, such that the patch **340** is visible using radiographic imaging.

(54) Additionally, the patch **340** can be used to secure the flap portions **369** after the flap portions **369** are pulled proximally away from the interior of the graft **361** and folded toward the outer surface of the graft **361**. For example, FIG. 5 is a close-up view of a portion of the graft **361** with the flap portions **369** attached to the patch **340**. As shown in FIG. 5, the flap portions **369** can be folded such that the flap portions **369** lie flat against the surface of the patch **340**, resulting in fenestration **363**. In some embodiments, the patch **340** can be coated in pressure-sensitive adhesive such that the flap portions **369** are secured to the patch **340** after being folded into contact with the patch **340**. In some embodiments, fasteners (not shown), such as sutures, staples, rivets, and micro-rivets, can be used to suture the flap portions **369** into a secure relationship with the patch **340** and/or the graft **361**. In some embodiments, the pressure-sensitive adhesive and/or fasteners can include a material with radiopaque properties such that the pressure-sensitive adhesive or fasteners are visible using radiographic imaging. In still other embodiments, the flap portions **369** can be bonded via thermal energy (e.g., welded) to the patch **340** such that the flap portions **369** are secured in the position shown in FIG. 5.

(55) Additionally, an optional outer patch can be coupled to the flap portions **369** and the patch **340** to further secure the flap portions **369** in place. For example, FIG. 6 is a close-up view of a portion of the graft **361** with an outer patch **342** shown as being transparent. The outer patch **342** can be coupled (e.g., adhered) to the patch **340** (shown in FIG. 5) and the flap portions **369** such that the flap portions **369** are secured between the patch **340** and the outer patch **342**. The outer patch **342** can include an adhesive on the side of the outer patch **342** in contact with the flap portions **369** and the patch **340**. In some embodiments, the outer patch **342** can be fastened (i.e., sutured, stapled, or riveted) to the flap portions **369**, the patch **340**, and/or the graft **361**. In some embodiments, the adhesive and/or fasteners can include a material with radiopaque properties such that the adhesive or fasteners are visible using radiographic imaging. In still other embodiments, the outer patch **342** can be bonded via thermal energy (e.g., welded) to the flap portions **369**, the patch **340**, and/or the graft **361**. Furthermore, the outer patch **342** can include radiopaque elements or be formed of a radiopaque material, similar to any of the patches described herein, such that the outer patch **342** is visible using radiographic imaging.

(56) Although not shown, in some embodiments the patch **340** could not be used. Instead, the flap portions **369** can be folded against the outer surface of the graft **361** and the outer patch **342** can be coupled to the graft **361** such that the flap portions **369** are sandwiched between the outer surface of the graft **361** and the outer patch **342**. The outer patch **342** can include adhesive on the side in contact with the graft **361** and the flap portions **369** to secure the outer patch **342** and the flap portions **369** in place. In some embodiments, the outer patch **342**, the flap portions **369**, and the graft **361** can be fastened (e.g., sutured, stapled, or riveted) in position. In some embodiments, the adhesive and/or fasteners can include a material with radiopaque properties such that the adhesive or fasteners are visible using radiographic imaging. In still other embodiments, the outer patch **342**, the flap portions **369**, and the graft **361** can be secured in position via welding.

(57) In some embodiments, a patch can include uniformly distributed radiopaque material. For example, FIG. 7 shows a front view of a patch **1340** secured to a graft **1361**. As shown in FIG. 7, the patch **1340** is substantially circular or donut-shaped (i.e., ring-shaped) and disposed on the graft **1361** such that the patch **1340** is concentric with a fenestration **1363** defined by the graft **1361**. The patch **1340** includes radiopaque material such that the patch **1340** can be visible via radiographic imaging. Due to the concentric positioning of the patch **1340** and the fenestration **1363**, the location of the fenestration **1363** can be identified via radiographic imaging during placement of the graft **1361** within a patient. For example, the fenestration **1363** can be aligned with a branch artery of a patient while using radiographic imaging to view the patch **1340**.

(58) In some embodiments, the patch **1340** can include polyurethane. The radiopaque material within the patch **1340** can be uniformly distributed and can include, for example, tungsten. In some embodiments, the patch **1340** can be disposed over stent struts **1392** of the graft **1361**. In some embodiments, the patch **1340** can be disposed such that the patch **1340** does not overlap the stent struts **1392**. In some embodiments, the patch **1340** can include cut-outs such that the patch **1340** does not overlap the stent struts **1392**. In some embodiments, the patch **1340** can be flexible. The patch **1340** can be secured to the graft **1361** via any suitable coupling method, and specifically via any suitable coupling method described herein. For example, the patch **1340** can be adhesively coupled to the graft **1361**. In some embodiments, the patch **1340** can be head bonded to the graft **1361**. In some embodiments, the patch **1340** can be sewn or otherwise fastened to the graft **1361**. Additionally, the patch **1340** can be applied to the graft **1361** before, simultaneously, or after the fenestration **1363** has been created, similarly as described above with reference to patch **340**.

(59) In some embodiments, a reinforcing member can include a first patch and a second patch joined to form a flexible grommet. The flexible grommet can be configured to sandwich one or more markers and/or one or more flap portions of a graft, such as an endograft, created through a fenestration process similarly as described above. In some embodiments, one or more markers can be placed on or near the flap portions. Each flap portion can be folded such that each flap portion sandwiches at least one of the markers between the flap portion and an outer surface of the graft. The first patch can be secured to the outer surface of the graft and the second patch can be secured to the inner surface of the graft such that the flap portions and/or markers are sandwiched between the first patch and the second patch. The first patch and the second patch can be secured to each other and/or the graft using, for example, one or more sutures or threads. Alternatively, the first patch and the second patch can be secured to each other and/or the graft via a sealing process such as heat sealing. In some embodiments, in an assembled configuration, the one or more markers can be disposed between the first patch and a flap portion and/or the graft or between the first patch and the second patch.

(60) FIG. **8** is a schematic illustration of a perspective view of a reinforcing member **500** (also referred to herein as a “flexible grommet”). The flexible grommet **500** includes a first patch **562** and a second patch **564**. The first patch **562** includes a first wire **566**. The first wire **566** can be a circular marker wire formed of a radiopaque material. The second patch **564** includes a second wire **568**. The second wire **568** can be a circular marker wire formed of radiopaque material. The second wire **568** can be movable between a pre-deployed or deployed, biased expanded configuration and an undeployed, compressed configuration for insertion through a fenestration of a graft. For example, the second wire **568** can be formed of a material having shape-memory properties, such as Nitinol. The second wire **568** can be secured (e.g., sewn or embedded) to the second patch **564** such as, for example, along, near, and/or concentric with the outer edge of the second patch **564** such that the shape and/or position of the second wire **568** can control the shape and/or position of the second patch **564**. Said another way, the second wire **568** can be configured to be compressed such that the second wire **568** and the second patch **564** have a smaller diameter in the undeployed configuration than the second wire **568** and the second patch **564** have in the expanded configuration. A surface of the first patch **562** and a surface of the second patch **564** can be heat bonded or joined through any suitable means, such as sewing, along seam **569**. In some embodiments, the first wire **566** and the first patch **562** can be the same or similar in structure and/or function to the second wire **568** and the second patch **564** such that either side of the flexible grommet **500** can be inserted through a fenestration in a graft and expand to a deployed configuration such that the flexible grommet **500** is secured within the fenestration and relative to the flexible graft.

(61) FIGS. **9A-9C** are schematic cross-sectional illustrations of the flexible grommet **500** in a pre-deployment configuration, a compressed configuration, and a deployed configuration, respectively. As shown in FIG. **9A**, the second wire **568** is in a biased expanded configuration such that the

second patch **564** is disc-shaped or donut-shaped (i.e., ring-shaped) prior to insertion of the flexible grommet **500** through a fenestration of a graft. FIG. **9B** shows the flexible grommet **500** in a compressed configuration in which the second wire **568** has been compressed or folded such that the second patch **564** can be inserted through a fenestration of a graft. FIG. **9C** shows the flexible grommet **500** in a deployed configuration in which the flexible grommet **500** has been inserted through a fenestration **563** in a graft wall **561** and is secured to the graft wall **561**. As shown in FIG. **9C**, after the second patch **564**, including the second wire **568**, has been inserted through the fenestration **563**, the second wire **568** can be allowed to automatically transition to the deployed configuration due to the second wire **568** having shape-memory properties. As a result of the second wire **568** expanding to the deployed configuration, the flexible grommet **500** is secured to the graft **561**. Specifically, the first patch **562** can be positioned on and/or engaged with a first side of the graft **561** and the second patch **564** can be positioned on and/or engaged with a second side of the graft **561**. In some embodiments, the first wire **566** and/or the second wire **568** can be configured to expand such that a patch-facing side of the first patch **562** and/or the second patch **564**, respectively, partially or fully engages with the surface of the graft **561** to minimize any space between the first patch **562** and/or the second patch **564** with the graft **561**. For example, the first wire **566** and/or the second wire **568** can be configured to expand such that the patch-facing side of the first patch **562** and/or the second patch **564** are disposed in contact with the graft **561** around or near at least the outer perimeter of the first patch **562** and/or the second patch **564**. Due to the first wire **566** and/or the second wire **568** having radiographic properties and surrounding the fenestration **563**, the graft **561** can be positioned within a patient such that the fenestration **563** is aligned with, for example, a branch artery, using radiographic imaging.

(62) In some embodiments, a reinforcing member can be formed by overmolding silicone or another elastomer, such as urethane, over fabric creating a composite material with a radiopaque ring embedded between the layers of silicone and the other elastomer. The reinforcing member can be formed as a flexible patch in a ring or donut-shape. For example, FIG. **10** is a side view of a graft **600** including a reinforcing member **670** (also referred to herein as a “patch”). The patch **670** can include a radiopaque ring or fiber **630** embedded within the patch **670**. The patch **670** is secured to the graft **600** via thread **650**. As shown, the radiopaque ring or fiber **630** can be arranged such that it is concentric with and/or surrounds a fenestration **663** in the graft **600** such that the location of the fenestration **663** can be identified using radiographic imaging.

(63) In some embodiments, a reinforcing member can be formed as a fabric grommet and encompass a radiopaque ring. FIG. **11** is a side view of a graft **700** including a reinforcing member **770** (also referred to herein as a “flexible grommet” or a “fabric grommet”). As shown in FIG. **11**, the flexible grommet **770** can be made of fabric and can include a first ring or donut-shaped fabric portion **771** and a second ring or donut-shaped fabric portion (not shown). The first donut-shaped fabric portion **771** and the second donut-shaped fabric portion can be attached along or near the inner surface of the first donut-shaped fabric portion **771** and the second donut-shaped fabric portion via thread **773**. Specifically, the first donut-shaped fabric portion **771** can be disposed on an exterior or first side of the graft **700** and the second donut-shaped fabric portion can be disposed on an interior or second side of the graft **700**. A radiopaque ring (not shown) can be disposed between the first donut-shaped fabric portion **771** and the second donut-shaped fabric portion, and can be secured in place relative to the first donut-shaped fabric portion **771**, the second donut-shaped fabric portion, and a fenestration **763** of the graft **700** by the thread **773**. In some embodiments, the radiopaque ring can be disposed between the first donut-shaped fabric portion **771** and the exterior or first side of the graft **700**. In some embodiments, the radiopaque ring can be disposed between the second donut-shaped fabric portion and the interior or second side of the graft **700**. Although thread **773** is shown and described, the first donut-shaped fabric portion **771** and the second donut-shaped fabric portion can be attached via any suitable means, such as via adhesive or heat bonding. Additionally, any suitable number of reinforcing members can be attached to the graft **700**. For

example, as shown in FIG. 11, the graft **700** can include a second flexible grommet **770A** secured in surrounding relation to a second fenestration **763A**. The second flexible grommet **770A** can be the same or similar in structure and/or function to the flexible grommet **770**. Thus, both the fenestration **763** and the second fenestration **763A** can be identified using radiographic imaging due to the radiographic ring surrounding each fenestration.

(64) In some embodiments, a reinforcing member can be formed as a fabric grommet and can include sutures of radiopaque thread. FIG. 12 is a perspective view of a reinforcing member **870** (also referred to herein as a “flexible grommet” or a “fabric grommet”). The flexible grommet **870** can be made of fabric, such as, for example, DACRON®, and can be formed as a first ring or donut-shaped fabric portion **871** and a second ring or donut-shaped fabric portion **872**. The first donut-shaped fabric portion **871** and the second donut-shaped fabric portion **872** can be attached along or near the inner surface of the first donut-shaped fabric portion **871** and the second donut-shaped fabric portion **872** via sutures **873**. The sutures **873** can be formed of radiopaque thread. Thus, the sutures **873** can be positioned on the flexible grommet **870** such that the sutures can indicate the location of a fenestration of a graft when the flexible grommet **870** is secured within the fenestration of the graft. The flexible grommet **870** can be attached to the graft via, for example, heat sealing or suturing. In some embodiments, the sutures **873** can be used both to secure the first donut-shaped fabric portion **871** to the second donut-shaped fabric portion **872** and to secure both the fabric portions **871**, **872** to a graft. In some embodiments, one or both of the first donut-shaped fabric portion **871** and the second donut-shaped fabric portion **872** can also include radiographic materials, such as having a radiographic substance uniformly distributed throughout, such that the fenestration to which the flexible grommet **870** is secured is more easily identifiable using radiographic imaging.

(65) In some embodiments, the reinforcing member can be formed as a substrate coupled to the graft such that one or more radiopaque elements are secured relative to a fenestration of the graft. For example, as shown in FIG. 13, a number of discrete radiopaque elements **1530** can be positioned in surrounding relation around a fenestration **1563** in a graft **1561**. The substrate **1540** can then be applied (e.g., overmolded) over the discrete radiopaque elements **1530** such that the discrete radiopaque elements **1530** are sandwiched between the substrate **1540** and the graft **1561**. Additionally, the substrate **1540** can be applied to the graft **1561** such that the area of the graft **1561** surrounding the fenestration **1563** is reinforced (e.g., fraying is reduced and/or prevented). Although seventeen discrete radiopaque elements **1530** are shown, any suitable number of radiopaque elements **1530** indicating the location of the fenestration **1563** can be used. In some embodiments, the radiopaque elements **1530** can be formed of or coated with, for example, gold, tantalum, and/or platinum. In some embodiments, the substrate **1540** can be formed of or include polyurethane.

(66) In some embodiments, rather than overmolding a substrate onto discrete radiopaque elements and a graft, a substrate can be overmolded onto a radiopaque coil ring and a graft. For example, FIG. 14 shows a radiopaque coil **1430** disposed in surrounding relation (e.g., concentrically disposed) around a fenestration **1463** defined in a graft **1461**. A radiopaque substrate **1440** can be overmolded over the radiopaque coil **1430** such that the radiopaque coil **1430** is sandwiched and/or embedded between the substrate **1440** and the graft **1461**. In some embodiments, the substrate **1440** can be formed of and/or include polyurethane. Due to the coil **1430** surrounding the fenestration **1463**, the location of the fenestration **1463** can be identified via radiographic imaging. In some embodiments the coil **1430** and/or the substrate **1440** can be disposed over stent struts **1492** of the graft **1461**. In some embodiments, the coil **1430** and/or the substrate **1440** can be disposed between and/or a distance from the stent struts **1492** such that the coil **1430** and/or the substrate **1440** do not overlap the stent struts **1492**. In some embodiments, the substrate **1440** can be formed of polyurethane and the graft **1461** can be formed of polyethylene terephthalate such that the application of thermal energy can bond the substrate **1440** to the graft **1461**.

(67) In some embodiments, the reinforcing member can be formed as a grommet formed of overmolded materials such that a radiopaque ring is embedded within the grommet. For example, a radiopaque ring can be embedded between layers of DACRON® and/or silicone via an overmolding process. In some embodiments, a flexible grommet can be formed by overmolding silicone or another elastomer, such as urethane, over fabric with a radiopaque ring embedded between the layers of silicone and the other elastomer.

(68) In some embodiments, a reinforcing member can be formed as a silicone grommet. For example, FIG. 15 is a side view of a graft **400** including a reinforcing member **470** (also referred to herein as a “leaf grommet”) in which a number of leaves are configured to be positioned on the inside of the graft **400** and a number of leaves are configured to be positioned on the outside of the graft **400**, thereby sandwiching the graft **400** between the leaves. The leaf grommet **470** includes one or more markers **430** embedded within the leaf grommet **470**. Although the one or more markers **430** are shown as discrete markers, in some embodiments the leaf grommet **470** can include a marker **430** formed as a radiopaque ring. The leaf grommet **470** can be secured to the graft **400** via one or more knots including a thread **450** and/or adhesive. Additionally, the leaf grommet **470** can have the same or similar structure and/or function as the leaf grommet **970** described below with reference to FIG. 16.

(69) FIG. 16 is a perspective view of a reinforcing member **970** (also referred to herein as a “grommet”). The grommet **970** can have the same or similar structure and/or function to the leaf grommet **470** described above with reference to FIG. 15. The grommet **970** can include a first portion **971** and a second portion **972**. The first portion **971** and the second portion **972** can be formed from silicone. The first portion **971** can include a donut or ring-shaped base **974a** and a number of projection portions **975a** extending from the base. Similarly, the second portion **972** can include a donut or ring-shaped base **974b** and a number of projection portions **975b** extending from the base. Although the first portion **971** and the second portion **972** are shown as including three projection portions each, the first portion **971** and the second portion **972** can include any suitable number of projection portions. The projection portions **975a** of the first portion **971** and the projection portions **975b** of the second portion **972** can be radially offset from each other such that projection portions **975b** can be positioned on the inside of a graft and the projection portions **975a** can be positioned on the outside of the graft, thereby sandwiching the graft and securing the grommet **970** in position relative to a fenestration of the graft.

(70) The grommet **970** can include a radiopaque ring **930** sandwiched between the first portion **971** and the second portion **972** such that the center of the grommet **970** (and thus, a fenestration of a graft to which the grommet **970** is attached) can be identified using radiographic imaging. In some embodiments, the silicone grommet can include radiopaque thread and/or discrete radiopaque beads, and/or can be formed with radiopaque materials. The grommet **970** can be attached to a graft via sutures, heat bonding, rivets, and/or any other suitable attachment means described herein. In some embodiments, the grommet **970** can reinforce the area of a graft surround a fenestration either before or after the fenestration is created. Although the grommet **970** is described as being formed from silicone, in some embodiments, the grommet can be formed of any other suitable material, such as an elastomer-polyurethane or polyamide. Additionally, although the grommet is shown as being shaped as having a ring-shaped base and projections, the grommet can be any suitable shape. For example, the grommet can be shaped similarly to an O-ring (i.e., having a ring-shaped base but no projections).

(71) In some embodiments, a reinforcing member can be formed as a grommet including a combination of fabric, a radiopaque element, and an elastomer material overmolded on the fabric. For example, FIG. 17 is a perspective view of a reinforcing member **1070** (also referred to herein as a “grommet”). The grommet **1070** can include a first portion **1071a**, a second portion **1071b**, and a third portion **1072**. The first portion **1071a** and the second portion **1071b** can be formed of, for example, silicone. The third portion **1072** can be formed of a fabric, such as, for example,

DACRON®. The first portion **1071a**, the second portion **1071b**, and the third portion **1072** can each be shaped as a circular disc with a central opening. The first portion **1071a** and the second portion **1071b** can be arranged to sandwich the third portion **1072** such that the openings of the first portion **1071a**, the second portion **1071b**, and the third portion **1072** are axially aligned. The grommet **1070** can include a first radiopaque element **1030A**, a second radiopaque element **1030B**, a third radiopaque element **1030C**, and a fourth radiopaque element **1030D** (collectively referred to herein as “the radiopaque elements **1030**”). Although four radiopaque elements **1030** are described and shown in FIG. **17**, any suitable number of radiopaque elements **1030** can be included. The radiopaque elements **1030** can include, for example, one or more markers of any suitable shape, such as a ring or bead. The radiopaque elements **1030** can be secured to the third portion **1072** via, for example, an adhesive or suture. In some embodiments, the radiopaque elements **1030** can be embedded between the first portion **1071a** and the third portion **1072** and/or between the second portion **1071b** and the third portion **1072**. For example, the elastomer material of the first portion **1071a** and/or the second portion **1071b** can be overmolded to the fabric of the third portion **1072** such that the radiopaque elements **1030** are embedded between the elastomer and the fabric. The grommet **1070** can be attached to a graft, such as an endograft, via, for example, heat sealing or suturing. The grommet **1070** can be attached to the graft such the first portion **1071A** and/or the third portion **1072** is directly secured to the graft and the second portion **1071b** is disposed within a fenestration of the graft. The second portion **1071b** can be shaped and sized to correspond to the shape and size of the fenestration of the graft such that the shape and size of the fenestration of the graft is maintained and/or to mark the shape and size of the fenestration of the graft. In some embodiments, the first portion **1071a** and/or the second portion **1071b** can include a radiopaque additive. For example, a radiopaque additive can be uniformly distributed throughout the first portion **1071a** and/or the second portion **1071b**.

(72) In some embodiments, a reinforcing member can be formed as a crimping apparatus used to mark and/or reinforce a fenestration in a graft. For example, FIG. **18** is a schematic illustration of a reinforcing member **1180** (also referred to herein as a “crimping apparatus”). The crimping apparatus **1180** can include a crimping structure **1182** and a wire or cable **1184**. The crimping structure **1182** can be formed from shim stock via, for example, laser cutting. The crimping structure **1182** can be crimpable or foldable around the wire **1184** such that the crimping structure **1182** can be crimped onto an area of a graft (e.g. a graft formed of DACRON® fabric) surrounding a fenestration. The wire **1184** can include a radiopaque material such that the wire **1184** can be visualized using radiographic imaging. The crimping structure **1182** can include outer tabs (e.g., a first outer tab **1186A** and a second outer tab **1186B** (collectively referred to herein as “outer tabs **1186**”)) and inner tabs (e.g., a first inner tab **1188A** and a second inner tab **1188B** (collectively referred to herein as “inner tabs **1188**”)). The inner tabs **1188** can be folded relative to the outer tabs **1186** through the fenestration upon application of the crimping apparatus to the graft such that the graft is secured between the outer tabs **1186** and the inner tabs **1188** and the wire **1184** surrounds and/or designates the location of the fenestration. The crimping apparatus **1180** can include any suitable number of outer tabs **1186** and inner tabs **1188**.

(73) In some embodiments, a reinforcing member can include an outer element and an inner element configured to be positioned on opposite sides of a fenestration in a graft and attached to each other through openings in the graft. The outer element can be formed of, for example, an elastomer material. The outer element can include discrete rigid connection features that can be formed of, for example, plastic. The inner element can be formed as a radiopaque metal ring and can include discrete rigid connection features corresponding to the connection features of the outer element. The connection features of the outer element and the connection features of the inner element can be aligned with the holes in the graft and attached to each other using, for example, ultrasonic welding. In some embodiments, the openings in the graft can be created at the same time the fenestration is created in the graft. In other embodiments, the openings can be created at the

time of attachment of the outer element and inner element to the graft.

(74) In some embodiments, a reinforcing member can include an outer element and an inner element configured to be positioned on opposite sides of a fenestration in a graft and attached to each other through openings in the graft via snap features. For example, FIG. 19A is a perspective view of a reinforcing member **1270** (also referred to herein as a “marking and reinforcing assembly” or an “assembly”). The assembly **1270** includes an outer element **1271** and an inner element **1272**. The outer element **1271** includes snap features **1276**. Although four snap features **1276** are shown, any suitable number of snap features can be included. The snap features **1276** can include a stem portion and a conical portion. The conical portion can have a larger diameter than the stem portion and can be tapered away from the stem portion. As shown in FIG. 19B, which is a perspective view of the outer element **1271**, the snap features **1276** can be shaped and sized such that the snap features **1276** can be inserted through an opening in another material, but cannot be withdrawn through the opening once inserted. The outer element **1271** can be formed of, for example, a flexible elastomer material. As shown in FIG. 19C, which is a perspective view of the inner element **1272**, the inner element **1272** can define openings **1277** shaped and sized such that the snap features **1276** can pass through the openings **1277** but cannot be withdrawn from the openings **1277** once fully engaged with the inner element **1272**. The inner element **1272** can be, for example, a laser cut shim. The inner element **1272** can be more rigid than the outer element **1271**. In some embodiments, the openings in the graft can be created at the same time the fenestration is created in the graft. In other embodiments, the openings can be created at the time of attachment of the outer element and inner element to the graft. In some embodiments, the inner element **1272** and/or the outer element **1271** can be formed of a radiopaque material or include discrete radiopaque elements, such as radiopaque beads or thread.

(75) In some embodiments, a method **1600** includes, at **1602**, generating a fenestration in a patient-specific prosthetic. The method **1600** can be used with any of the reinforcement members described herein. The fenestration can correspond to a location of a branch blood vessel in a portion of a patient's blood vessel. The method **1600** can further include coupling a marker to the patient-specific prosthetic, at **1604**. The marker can include a member configured to be secured to the patient-specific prosthetic such that the patch surrounds the fenestration. The marker can also include at least one radiopaque element configured to indicate the location of the fenestration via radiographic imaging. In some embodiments, the coupling of the marker to the patient-specific prosthetic is performed prior to the generating of the fenestration. In some embodiments, the coupling of the marker to the patient-specific prosthetic is performed after the generating of the fenestration. In some embodiments, the coupling includes applying thermal energy to the marker and the patient-specific prosthetic. In some embodiments, the marker and the patient-specific prosthetic both include polyethylene terephthalate. In some embodiments, the coupling includes overmolding the member onto the at least one radiopaque element and the patient-specific prosthetic. In some embodiments, the coupling of the marker to the patient-specific prosthetic is performed simultaneously to the generating of the fenestration. Optionally, the method **1600** can further include positioning the patient-specific prosthetic within a patient such that the marker aligns with the location of the branch blood vessel using radiographic imaging, at **1606**.

(76) While various embodiments of the system, methods and devices have been described above, it should be understood that they have been presented by way of example only, and not limitation. Where methods and steps described above indicate certain events occurring in certain order, those of ordinary skill in the art having the benefit of this disclosure would recognize that the ordering of certain steps may be modified and such modifications are in accordance with the variations of the invention. Additionally, certain of the steps may be performed concurrently in a parallel process when possible, as well as performed sequentially as described above. The embodiments have been particularly shown and described, but it will be understood that various changes in form and details may be made.

(77) For example, although various embodiments have been described as having particular features and/or combinations of components, other embodiments are possible having any combination or sub-combination of any features and/or components from any of the embodiments described herein. In addition, the specific configurations of the various components can also be varied. For example, the size and specific shape of the various components can be different than the embodiments shown, while still providing the functions as described herein.

Claims

1. A system, comprising: a) a graft (361), the graft having an interior and an outer surface, and includes flap portions (369) formed in a side of the graft (361) defining a pilot hole (365) and cuts (367); and b) a ring-shaped patch (340) that defines an internal edge and is coupled to an area of the graft surrounding the flap portions; c) wherein the graft is an endograft; and d) an outer patch that is coupled to the flap portions and to the ring-shaped patch, thereby further securing the flap portions in place.
2. The system of claim 1, wherein the graft includes eight flap portions, thereby defining eight cuts.
3. The system of claim 1, wherein the edges of the cuts are aligned with the internal edge of the ring-shaped patch.
4. The system of claim 1, wherein the ring-shaped patch is formed of a radiopaque material.
5. The system of claim 1, wherein the ring-shaped patch is coupled to the graft by an adhesive, sutures or welding.
6. The system of claim 1, wherein the flap portions are pulled proximally away from the interior of the graft and folded toward the outer surface of the graft.
7. The system of claim 6, wherein the flap portions are attached to the ring-shaped patch.
8. The system of claim 7, further including a plurality of fasteners, whereby the flap portions are secured to the ring-shaped patch.
9. The system of claim 6, further including fasteners, whereby the flap portions are attached to the graft.
10. The system of claim 9, wherein the fasteners include at least one of sutures, staples, rivets or microrivets.
11. The system of claim 1, further including an adhesive on a side of the outer patch that is in contact with the flap portions and the ring-shaped patch, and that secures the ring-shaped patch to the outer patch.
12. The system of claim 1, further including fasteners that fasten the outer patch to the flap portions and to the ring-shaped patch.
13. The system of claim 12, wherein the fasteners include at least one of sutures, staples, and rivets.
14. The system of claim 1, further including fasteners, wherein the outer patch is fastened to at least one of flap portions, the patch, and the graft.
15. The system of claim 14, wherein the fasteners include at least one of sutures, staples, and rivets.
16. The system of claim 1, further including fasteners, wherein the outer patch is fastened to the flap portions, the ring-shaped patch, and the graft.
17. The system of claim 16, wherein the fasteners include at least one of sutures, staples, and rivets.
18. A system, comprising: a) a graft (361), the graft having an interior and an outer surface, and includes flap portions (369) defining a pilot hole (365) and cuts (367); and b) a ring-shaped patch (340) that defines an internal edge and is coupled to an area of the graft surrounding the flap portions; wherein flap portions are attached to the ring-shaped patch and the flap portions lie flat against a surface of a ring-shaped patch, thereby forming a fenestration.
19. The system of claim 18, wherein the ring-shaped patch is coated in pressure-sensitive adhesive, such that the flap portions are secured to the ring-shaped patch.
20. The system of claim 18, wherein the fasteners include at least one of sutures, staples, rivets or

micro-rivets.

21. The system of claim 18, wherein the fasteners include a material with radiopaque properties.
 22. The system of claim 18, wherein the flap portions are welded to the ring-shaped patch.
 23. A system, comprising: a) a graft (361), the graft having an outer surface, and includes flap portions (369) defining a pilot hole (365) and cuts (367), wherein the flap portions are folded against the outer surface of the graft; and b) an outer patch (342) that is coupled to the graft such that the flap portions are sandwiched between the outer surface of the graft and the outer patch.
 24. The system of claim 23, wherein the graft is an endograft.
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